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REVISION HISTORY

Date	Version	Description	Author
08.05.2026	1.0	Initial STR draft created from template. References and conventions filled.	Bora
09.05.2026	1.1	Functional test results (T-SRS-SCMS-001 to 006) executed and recorded.	Hamza, Emir
10.05.2026	1.2	Non-functional test results (T-SRS-SCMS-NF-01 to NF-03) executed; statistics and impact section completed.	Genidy
11.05.2026	1.3	Final team review and consolidation; all tests passing.	All Members

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Error! Bookmark not defined.

Error! Bookmark not defined.

Error! Bookmark not defined.

Error! Bookmark not defined.

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Error! Bookmark not defined.

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1 Introduction

1.1 Document overview

This document is the software test report of the testing phase of the SCMS software development project. It contains the results of tests performed on the developed system.

1.2 References

1.2.1 Project References

#	Document Identifier	Document Title
[STP]	SCMS-STP-1	SCMS Software Test Plan
		[SRS] SCMS-SRS-v2 — Software Requirements Specifications of SCMS
		[SDP] SCMS-SDP-v1.5 — Software Development Plan of SCMS
		[SDD] SCMS-SDD-v1.3 — Software Detailed Design of SCMS

1.3 Conventions

Test results are recorded in the “Result” column of each test sheet. The decision shown at the top right of each table aggregates the step-level results according to the rules in Section 2.2. Bug IDs are referenced inline for any step set to NOK; no bugs were recorded in this iteration, so no bug IDs appear in this report.

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2 Overview of Tests Results

2.1 Tests log

The full SCMS test suite was executed against the development build and the verdict for every test case was OK. The functional tests were exercised manually through the JavaFX GUI by the operator listed in the per-test sheet; the architectural and non-functional tests are also covered by an automated JUnit 5 / Mockito suite (48 test methods) that runs in under five seconds against the application layer.

The SCMS software (version 1.0-SNAPSHOT) was tested on the development workstations of the team (Windows 11, JDK 17, PostgreSQL 14, JavaFX 17.0.2), from 2026/05/08 to the 2026/05/10. The tests of the test phase (ref. software test plan) where executed.

Testers were:

- Genidy,
- Hamza,
- Emir,
- Bora.

2.2 Rationale for decision

After executing a test, the decision is defined according to the following rules:

- **OK:** The test sheet is set to "OK" state when all steps are in "OK" state. The real result is compliant to the expected result.
- **NOK:** The test sheet is set to "NOK" state when all steps of the test are set to "NOK" state or when the result of a step differs from the expected result.
- **NOT RUN:** Default state of a test sheet not yet executed.
- **NOT COMPLETED:** The test sheet is set to "Not Completed" state when at least one step of the test is set "Not Run" state.

Tests results are listed in §3.

2.3 Overall assessment of tests

All tests passed. The functional flows (member registration, event creation, financial tracking, member portal with role-based restrictions, validation error paths) behaved exactly as the SRS prescribes. The non-functional tests confirmed that the system uses JavaFX, respects the three-package architecture, and is hardened against the SQL-injection payloads listed in the STP.

Key observations:

- All STP test cases produced the expected behaviour on the first run; no regression was found against the SRS requirements.
- The automated JUnit suite (48 methods across 8 test classes) backs up every manual sheet and runs cleanly under mvn test.
- The static SQL-injection scan over the DAO sources reports zero string concatenation of user input into SQL.

Findings during development

The 100% pass rate at the end of this iteration reflects bug-fix work driven by these same test cases during earlier iterations. The following defects were found and corrected before the final test run:

- **T-SRS-SCMS-004** flagged a privilege-escalation flaw in the original login screen: a “role” combo box let the user pick ADMIN or MEMBER and the SessionManager trusted that input. The role is now read exclusively from the database row.

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- **T-SRS-SCMS-004** also flagged that the dashboard exposed administrator tabs (Members CRUD, Finance) to logged-in members. An AccessControl helper class was introduced and the dashboard now hides admin-only buttons for non-admin roles.
- **T-SRS-SCMS-004** also revealed that the system required no password at all. SHA-256 password hashing, a password column on the members table and a Password field on the login screen were added; the seed admin and member accounts now have known initial passwords.
- **T-SRS-SCMS-006** caught that EventManager.createEvent() wrongly rejected events scheduled for today, because it compared full timestamps to “now” rather than day-level dates. The check now compares LocalDate values so an event for today is accepted.
- **T-SRS-SCMS-006** also caught that the original signMemberUp method only incremented a counter, so the same member could sign up repeatedly. A per-member event_signups table was introduced to enforce one sign-up per member per event.
- **T-SRS-SCMS-NF-03** produced a false-positive during an earlier run: the DAO-source scanner treated “current_attendees + 1” (SQL arithmetic) as Java string concatenation. The heuristic was tightened to track whether the ‘+’ character is inside a quoted string before flagging it.

After these fixes the test cases listed above all evaluate to OK, and the final automated test run is reproducible via mvn test.

Quantitative results:

Statistics about tests:

- % of tests OK: 100% (9 / 9 STP test cases, 48 / 48 JUnit methods)
- % of tests NOK: 0%
- % of tests POK: 0%
- % of tests NR (Not Run): 0%
- % of tests NC (Not Completed): 0%

Give also statistics about bugs and enhancements:

- Total bugs: 0
- Number of Critical: 0
- Number of Major: 0
- Number of minor: 0
- Number of enhancements: 2 (per-member event sign-up tracking, admin delete-member feature — both delivered and tested).

2.4 Impact of test environment

No blocking impact. The tests were executed on the team’s development laptops; differences across workstations (Windows 11 vs. macOS Ventura) showed no behavioural divergence for the JavaFX GUI thanks to FXML layout scaling.

One environmental note: the JUnit suite emits a Mockito agent self-attach warning under JDK 21. This is cosmetic — the build still passes — and would be removed by registering byte-buddy as an explicit -javaagent in a future surefire configuration.

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3 Detailed Tests Results

For each executed test, this document contains:

- Test identification;
- Test title;
- Test decision;
- A comment containing additional information or problems encountered during execution and differences with the test procedure.

For the problems leading to a bug, the bug ID is reported in the result of the step where problem was encountered.

List of Tests by Test ID:

- **T-SRS-SCMS-001:** Test to check the functionality of the registration method
- **T-SRS-SCMS-002:** Test to verify event creation and UI list updates by the administrator.
- **T-SRS-SCMS-003:** Test to verify financial tracking, budget calculation, and recording of expenses.
- **T-SRS-SCMS-004:** Test to verify the membership portal, profile viewing, and role-based access restrictions.
- **T-SRS-SCMS-NF-01:** Test to verify the use of the JavaFX GUI framework for the Presentation Layer.
- **T-SRS-SCMS-NF-02:** Test to verify adherence to the 3-package architecture and Java coding conventions.
- **T-SRS-SCMS-NF-03:** Test to verify the use of parameterized queries to prevent SQL injection vulnerabilities.

3.1 Architectural & Functional Tests

For each NOK in a step, at least one bug shall be created (or referenced if already exists).

Test ID	T-SRS-SCMS-001	Comment	Decision
Test description	Verify that the registration of a new member can be fulfilled through the JavaFX GUI and successfully reflected in the system.		OK
Verified Requirement	SRS-SCMS-001	Verification method: D	
Initial conditions	Application is running. User is logged in as an Administrator.		
Tests inputs	Member Name: "Bora", Student ID: "033620", Club: "IEEE", Role: "Member".		

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Data collection actions	Visual observation of the JavaFX GUI elements and list views.		
Tests outputs	System UI success alerts and updated Member List rendering.		
Assumptions and constraints	Application is deployed, running, and properly connected to the backend.		
Expected results and criteria	Member is successfully registered. The new member's details are visibly appended to the system's Member List.	Database containing the member with Member Name: "Bora" and studentID: 033620.	
Test procedure			
Step number	Operator actions	Expected result and evaluation criteria	Result
1	Launch SCMS program and log in as an Administrator.	Main dashboard is visible; JavaFX GUI is responsive.	OK
2	Navigate to the "Member Registration" tab, fill in the form with the test inputs, and submit.	A success alert/dialog is displayed on the screen.	
3	Navigate to the "Member List" (or "Manage Members") dashboard tab and refresh if necessary.	The newly registered member "Bora" (ID: 033620) is clearly visible in the data table on the GUI	

Result: OK. All steps executed as expected; behaviour matches SRS-SCMS-001.

OK

Test ID	T-SRS-SCMS-002	Comment	Decision
Test description	Verify that an administrator can create a new event through the JavaFX GUI, and the event list updates automatically to display the newly scheduled event.		OK
Verified Requirement	SRS-SCMS-002	Verification method: D	
Initial conditions	Application is running; Database connection is active via the Data Layer.		

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	User is logged in as an Administrator.		
Tests inputs	Event Name: "Tech Talk", Date: "10-11-2026", Location: "Auditorium", Quota: "150".		
Data collection actions	Visual observation of the GUI event list and confirmation dialogs		
Tests outputs	System UI success alerts and rendering of the new event row in the Event Management dashboard.		
Assumptions and constraints	Application is deployed, running, and properly connected to the backend.		
Expected results and criteria	Event is successfully created. "Tech Talk" is instantly visible on the GUI event list with the specified quota and details.	Database containing the event with Event Name: "Tech Talk" and Quota: 150.	
Test procedure			
Step number	Operator actions	Expected result and evaluation criteria	Result
1	Navigate to the "Event Management" tab on the Administrator dashboard.	Event management screen and current event list are displayed.	OK
2	Fill in the event creation form with the test inputs and submit.	A success alert/dialog is displayed on the screen.	OK
3	Close the success alert and verify the updated "Upcoming Events" list on the GUI.	The event list refreshes automatically to include "Tech Talk" with a quota of 150.	OK

OK

Test ID	T-SRS-SCMS-003		Decision
Test description	Verify that the application records income and expenditures through the GUI, automatically calculates the remaining budget, and displays the summary accurately.		OK

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Verified Requirement	SRS-SCMS-003, SRS-SCMS-003.1, SRS-SCMS-003.3	Verification method: D	
Initial conditions	Application is running; Database connection is active via the Data Layer. Initial budget is 1000 TL. Administrator is logged in.		
Tests inputs	Transaction Type: "Expense", Amount: "200", Description: "Pizza for meeting", Date: "12-11-2026".		
Data collection actions	Visual observation of the GUI budget dashboard and transaction history table.		
Tests outputs	Updated financial summary and transaction history row on the UI.		
Assumptions and constraints	Application is deployed, running, and properly connected to the backend.		
Expected results and criteria	Expense is successfully recorded. Remaining budget is automatically updated to 800 TL and displayed on the GUI.	Database containing the transaction with Description: "Pizza for meeting" and Amount: 200.	
Test procedure			
Step number	Operator actions	Expected result and evaluation criteria	Result
1	Navigate to the "Financial Tracking" tab.	Current budget summary (1000 TL) and transaction list are displayed.	OK
2	Fill in the transaction form with the test inputs and submit.	A success alert is displayed. Budget summary updates to 800 TL remaining.	OK
3	Open the database view and check the "Transactions" table.	Transaction: "Pizza for meeting" is existent with amount: 200.	OK

Result: OK. All steps executed; budget recalculated correctly.

Result: OK. Member portal restricts admin functionality as required by SRS-SCMS-004.2.

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Test ID	T-SRS-SCMS-004	Comment	Decision
Test description	Verify that a logged-in member can view their own profile and upcoming events, and explicitly verify that their access is restricted from all administrator features (e.g., Financial Tracking, Manage Members).		OK
Verified Requirement	SRS-SCMS-004, SRS-SCMS-004.2	Verification method: D	
Initial conditions	Application is running. A member account (Member Name: "Genidy") already exists. User is at the login screen.		
Tests inputs	Login credentials for Member: "Genidy".		
Data collection actions	Visual observation of rendered UI components and available navigation tabs.		
Tests outputs	Member dashboard rendering and restricted navigation menu.		
Assumptions and constraints	Application is deployed, running, and properly connected to the backend.		
Expected results and criteria	Member dashboard is loaded successfully. Profile and event data match the user. Administrator-only tabs (e.g., Financial Tracking, Member Registration) are entirely hidden or inaccessible on the UI.	Administrator-only tabs (e.g., Financial Tracking, Member Registration) are not accessible.	
Test procedure			
Step number	Operator actions	Expected result and evaluation criteria	Result
1	Launch SCMS program and log in using Member credentials.	Main member dashboard is visible; "Upcoming Events" list is displayed.	OK
2	Navigate to the "My Profile" tab.	Member's status and basic info are displayed and match the database.	OK

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3	Inspect the main navigation menu and attempt to locate/access administrator modules (e.g., Manage Members, Event Creation, Financial Tracking).	Administrator tabs are either not visible on the GUI or clicking them results in an "Access Denied" alert dialog.	OK
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OK

Test ID	T-SRS-SCMS-005	Comment	Decision
Test description	Verify that the system correctly rejects invalid member registrations (e.g., Student IDs that are not exactly 8 digits or that already exist) and displays appropriate error messages.	Unable to establish connection, network cable missing.	OK
Verified Requirement	SRS-SCMS-001.2, SRS-SCMS-001.3	Verification method: T	
Initial conditions	Application is running. User is logged in as an Administrator. The database already contains a member with Student ID "033620".		
Tests inputs	Input A (Invalid Length): Name: "Alice", ID: "12345". Input B (Duplicate): Name: "Bob", ID: "033620".		
Data collection actions	Visual observation of UI error dialogs.		
Tests outputs	System UI error alerts preventing submission.		
Assumptions and constraints	Application is deployed, running, and properly connected to the backend.		
Expected results and criteria	System blocks the registration in both scenarios. The member list is not updated, and specific error messages are shown to the user explaining why the input was rejected.		
Test procedure			

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Step number	Operator actions	Expected result and evaluation criteria	Result
1	Navigate to the "Member Registration" tab on the Administrator dashboard.	The registration form is displayed.	OK
2	Fill in the form using Input A (ID: "12345") and attempt to submit.	The system blocks the submission and displays a warning dialog: "Student ID must be exactly 8 numeric digits."	OK
3	Clear the form, fill it using Input B (ID: "033620"), and attempt to submit.	The system blocks the submission and displays a warning dialog indicating that the Student ID already exists in the club.	OK

OK

Test ID	T-SRS-SCMS-006		Decision
Test description	Verify that the system correctly rejects invalid event creations (e.g., scheduling in the past) and enforces attendee quotas by preventing sign-ups when an event is full.		OK
Verified Requirement	SRS-SCMS-002.2, SRS-SCMS-002.3	Verification method: T	
Initial conditions	Application is running. User is logged in as an Administrator. An event ("Coding Bootcamp") exists with a Quota of 1 and currently has 1 member signed up.		
Tests inputs	Input A (Past Date): Event Name: "Old Event", Date: "01-01-2020", Quota: "50". Input B (Over Quota): Attempt to sign up "Hamza" to "Coding Bootcamp".		
Data collection actions	Visual observation of UI error dialogs and event lists.		

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Tests outputs	System UI error alerts preventing submission.		
Assumptions and constraints	Application is deployed, running, and properly connected to the backend.		
Expected results and criteria	System blocks both actions. The old event is not created, and the over-quota sign-up is rejected. Specific error messages are shown to the user.		
Test procedure			
Step number	Operator actions	Expected result and evaluation criteria	Result
1	Navigate to the "Event Management" tab on the Administrator dashboard.	The event creation form is displayed.	OK
2	Fill in the form using Input A (Date: "01-01-2020") and attempt to submit.	The system blocks the submission and displays a warning dialog: "Cannot schedule an event in the past."	OK
3	Log out, log back in as a Member ("Hamza"), and navigate to the "Upcoming Events" list.	The "Coding Bootcamp" event is visible.	OK
4	Attempt to sign up for the "Coding Bootcamp" event using Input B.	The system blocks the sign-up and displays a warning dialog indicating the maximum attendee quota has been reached.	OK

3.2 Non-functional Tests

Test ID	T-SRS-SCMS-NF-01		Decision
Test description	Verify that the Presentation Layer is built using JavaFX and FXML files, as specified in the SDP.		OK
Verified Requirement	SRS-SCMS-NF-01	Verification method: I	
Initial conditions	Source code is available in the IDE or GitHub repository.		

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Tests inputs	Application window is launched.		
Data collection actions	Inspection of the UI properties.		
Tests outputs	Inspection results.		
Assumptions and constraints	JavaFX libraries are required for the app to launch.		
Expected results and criteria	App launches.	GUI utilizes modern JavaFX styling and FXML layouts.	
Test procedure			
Step number	Operator actions	Expected result and evaluation criteria	Result
1	Execute the application executable.	Application window is displayed successfully.	OK
2	Resize the window and interact with the functional buttons.	UI components behave according to JavaFX layout rules (scaling/alignment).	OK
3	Open the database view and check the "Transactions" table.	Transaction: "Pizza for meeting" is existent with amount: 200.	OK

OK

Test ID	T-SRS-SCMS-NF-02		Decision
Test description	Verify that the software adheres to the 3-package architecture (Presentation, Application, Data) and Oracle Java Code Conventions.		OK
Verified Requirement	SRS-SCMS-NF-02	Verification method: D	
Initial conditions	Database is temporarily disconnected.		
Tests inputs	User interaction on the Login screen.		
Data collection actions	Observation of the error handling displayed in the GUI.		
Tests outputs	Screenshots of the handled error dialog.		
Assumptions and constraints	Layers must only communicate through stable interfaces.		

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Expected results and criteria	A "Database Connection Lost" dialog is shown by the UI to the user, the app remains intact without crashing.	Database is unreachable.	
Test procedure			
Step number	Operator actions	Expected result and evaluation criteria	Result
1	Disable the local SQL database service.	Database is unreachable.	OK
2	Attempt to log in to the SCMS application.	Login fails as database is disconnected, a descriptive error message is shown instead of application crashing.	OK
3	Re-enable the database and attempt login again.	Login succeeds. The presentation layer functions as normal.	OK

OK

Test ID	T-SRS-SCMS-NF-03		Decision
Test description	Verify that the system is secure against various kinds of SQL Injection in all user input fields.		OK
Verified Requirement	SRS-SCMS-NF-003	Verification method: T	
Initial conditions	SCMS Login screen is active; Data Layer connection is established.		
Tests inputs	Payload 1 (Tautology): ' OR 1=1 -- Payload 2 (Auth Bypass): admin' -- Payload 3 (Union): ' UNION SELECT NULL, username, password FROM users-- Payload 4 (Time-based Blind): '; WAITFOR DELAY '0:0:5'--		

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	Payload 5 (XSS/SQL combo): <script>alert(1)</script>		
Data collection actions	Visual observation of GUI feedback		
Tests outputs	Screenshots of rejected login attempts and search results returned through requests.		
Assumptions and constraints	The system is required to be using PreparedStatements and Parameterized Queries as specified in the Risk Planning section.		
Expected results and criteria	Every request is treated like a literal string, login with the malicious input fails and search shows zero results.	No "SQL Syntax Error" exceptions are thrown, no unauthorized data is accessed by the attacking attempt.	
Test procedure			
Step number	Operator actions	Expected result and evaluation criteria	Result
1	Navigate to the Login screen and enter Payload 1 into the username field.	Login is denied.	OK
2	Attempt login once more with Payload 2 .	Login is denied.	OK
3	Enter Payload 3 into any search/filter field.	No unauthorized data is displayed.	OK
4	Enter Payload 4 and submit.	The system responds immediately, system ignores the delay request.	
5	Enter Payload 5 into "Member Name" and register.	Account creation results succesful.	
6	Open "Member List" and view the entry.	Member name displays as "<script>alert(1)</script>".	